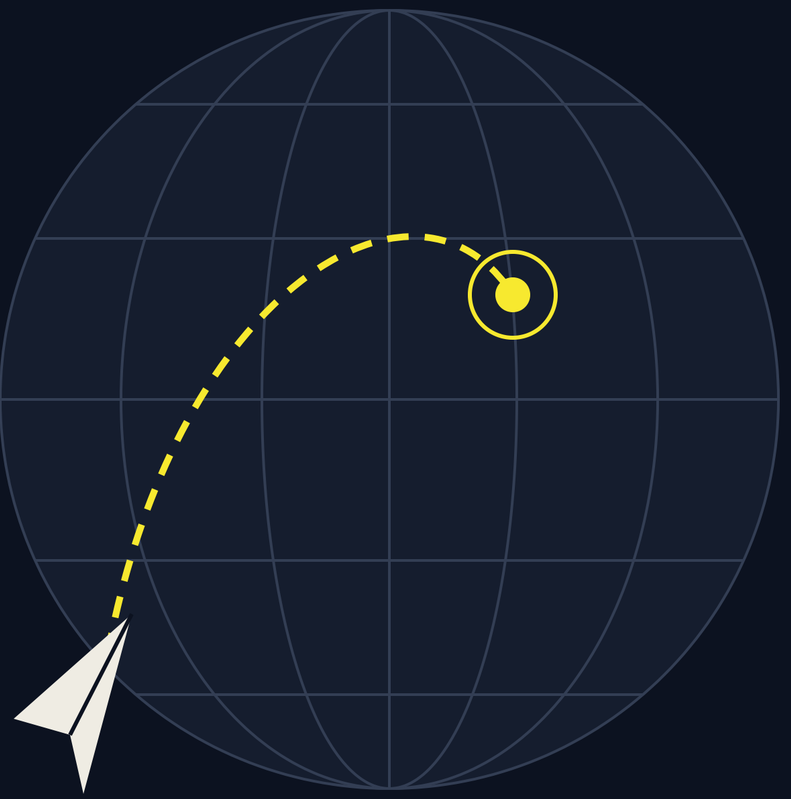
LAYOVER 2026 · TINKERHUB × THE VISA GUY

VISA ROULETTE



AI sees. Rules decide.
Every verdict cites its source.

TINKERSPACE CALICUT

10-11 OCT 2026

RYYAN SAFAR + FATHIMA

Flight ticket says go. Passport says no.

Lakshmi Nair, a specimen traveller from our demo, reaches the gate. Her passport expires 10 Mar 2027. Her flight leaves 11 Oct 2026.

Many countries want six months left on a passport. She would find out at the desk.

PASSPORT VALIDITY LEFT

4 months 27 days

RULE: 6 MONTHS



Throw a paper plane at a globe. It lands where your passport can go.

AI sees. Rules decide. Every verdict cites its source.

HOW IT WORKS

Scan. Throw. Land.

STEP 1

Scan

Show a passport and a flight ticket to the camera. OCR reads the passport, the IATA barcode gives the flight, and a checklist ticks live.

STEP 2

Throw

Throw a paper plane fitted with an ESP32 and an IMU. A webcam flick or a single switch works too. The globe slows and lands.

STEP 3

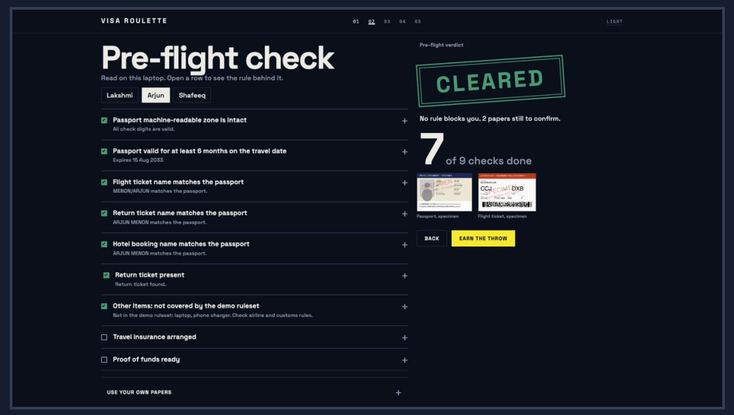
Land

Get the visa verdict with its source, the documents to carry, destination highlights, and a QR trip card for your phone.

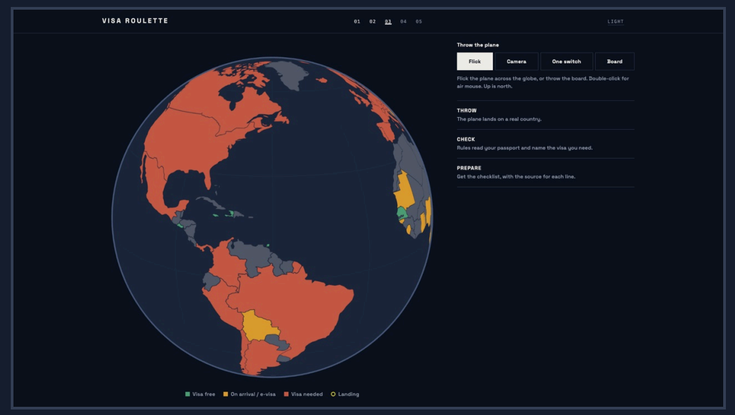
PLUS: OFFICER INTERVIEW · PASSPORT DUEL · WRONG-DOCUMENT PICK · SOUVENIR WALL

Live demo.

The 90-second booth loop. No internet needed.



1 · Pre-flight check



2 · Throw the plane



3 · Landing verdict

AI sees. Rules decide.

AI SEES

Camera

Passport and flight ticket photographed at the desk.

AI SEES

OCR + vision

Tesseract reads the MRZ, zxing reads the barcode. Check digits catch bad reads.

RULES DECIDE

Rules engine

JSON rule files. Same input, same answer. No model in the loop.

THE VERDICT

Cited

Status, documents, source link and as-of date on every line.

A local LLM may explain a verdict in plain words. It cannot change one.

Destination unknown? The booth says unknown. It never guesses.

DASHED = PROBABILISTIC · SOLID = DETERMINISTIC

The plane is the controller.

±16 g
ACCEL RANGE

500 Hz
SAMPLE RATE

±2000 °/s
GYRO RANGE

< 7 m/s²
RELEASE THRESHOLD

ESP32 + MPU-6050 inside a paper plane, streaming over WiFi to the laptop.

Launch push at 20 m/s² or more. Release when the reading drops below 7 m/s² for 20 ms.

Bad throws are rejected with a reason: no rest, too weak, mount angle, straight up.

No sensor? A webcam flick or a single Space key flies the same model.

Thank you.

TRY IT **layoverhackathon.vercel.app**

CODE **github.com/ryyansafar/visa-roulette**

WHAT WE BUILT

One night, one repo.

BACKEND	FastAPI, Python
READING	Tesseract MRZ, zxing IATA barcode, OpenCV
JUDGING	JSON rules with source, as-of date, verified flag
FRONT END	Vanilla ES modules, three.js r170 globe
HARDWARE	ESP32 + MPU-6050 over WebSocket
LOCAL AI	Ollama gemma3n:e4b, faster-whisper voice

320+
TEST FUNCTIONS

113
DESTINATIONS RATED, INDIAN PASSPORT

11
CURATED DESTINATION PAGES

0
INTERNET NEEDED FOR THE CORE FLOW

Said plainly.

HONEST LIMITS

Demo rules are unverified. Every verdict says so.

The hosted version has sample travellers only: no hardware, no live scans.

Arbitrary photo scans work on the local booth only.

Four passports covered; Indian is the deepest.

WHAT IS NEXT

Verify every rule against official sources.

Pros-and-cons reports per destination.

More accessible inputs beyond mouse and camera.

WebMCP tools so an agent can ask the booth.

Hard questions, honest answers.

Is the AI deciding visas?

No. Rules decide. AI only reads and explains.

How accurate is the rules data?

Demo rules, unverified, dated 2026-10-10. Cited.

What if OCR fails?

Check digits flag it. Retake hint, or a sample traveller.

Does it work offline?

Core flow yes. Local model, voice, QR hand-off.

Real passports? Privacy?

Sample or fake only. Data stays on the laptop.

What if the hardware dies?

Webcam flick, Space key, or the video.